

INTERNSHIP REPORT

ON

Smart PIR Motion Sensor Security
Automation System Using
Arduino UNO

(TASK-3)

IN THE DOMAIN OF

EMBEDDED SYSTEMS AND IOT

BY

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ABSTRACT

This project demonstrates a PIR (Passive Infrared) Motion Sensor-based Security System implemented using a microcontroller in the Wokwi simulation environment. The system detects human motion by sensing changes in infrared radiation and triggers an alert using LED and buzzer. The main objective is to design a low-cost and efficient security system using embedded systems concepts. The simulation verifies the working of motion detection and alert mechanism, making it suitable for real-time security applications such as homes, offices, and industries.

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1. INTRODUCTION

A Smart PIR Motion Sensor Security Automation System is an embedded system designed to automatically detect human movement and provide security monitoring. Motion detection systems are widely used in homes, offices, and industrial environments for improving safety and preventing unauthorized access. PIR sensors are commonly used because they can detect changes in infrared radiation emitted by the human body.

In this project, Arduino UNO is used as the main controller to process signals received from the PIR motion sensor. When motion is detected, the system automatically activates the LED and displays monitoring messages through the Serial Monitor. The project demonstrates automation logic, sensor interfacing, and real-time monitoring using the Wokwi simulation platform.

2. OBJECTIVES

- To design a smart motion detection security automation system using Arduino UNO
- To detect human movement using a PIR motion sensor.
- To implement automation logic using IF and ELSE conditions.
- To automatically control output devices based on sensor input.

3. COMPONENT USED

A PIR Motion Sensor Security System is used to detect human movement and provide security alerts. In this project, Arduino UNO is used as the microcontroller to control the system operations. The PIR sensor detects motion by sensing infrared radiation changes from the human body. An LED is used as an indicator to show motion detection, and a resistor is connected to protect the LED by limiting the current flow. The project was simulated using the Wokwi platform to understand the working of embedded systems and motion detection applications. The following Components are:

(i)Arduino

Arduino UNO is the main microcontroller used in this project to control the complete motion detection system. It receives input signals from the PIR motion sensor and processes them according to the programmed instructions. Based on the sensor output, the Arduino UNO activates the LED indicator when motion is detected. It is built on the ATmega328P microcontroller and is widely used in embedded systems and automation projects due to its simplicity and ease of programming.



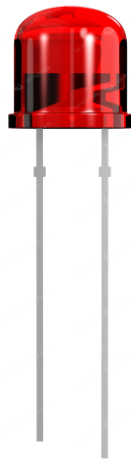
(ii) PIR Motion Sensor

The PIR (Passive Infrared) Motion Sensor is used to detect human movement in the surrounding area. It works by sensing changes in infrared radiation emitted by living bodies such as humans. Whenever a person moves within the detection range, the sensor generates a signal and sends it to the Arduino UNO. PIR sensors are commonly used in security systems because they consume low power and provide reliable motion detection. In this project, the PIR sensor acts as the main sensing device for detecting intrusion or movement.



(iii) LED

The LED (Light Emitting Diode) is used as an output indicator in this project. It glows whenever the PIR sensor detects motion and sends a signal to the Arduino UNO. LEDs are commonly used in electronic circuits because they consume less power and respond quickly. In this project, the LED acts as a simple alert device for security indication.



(iv) Resistor

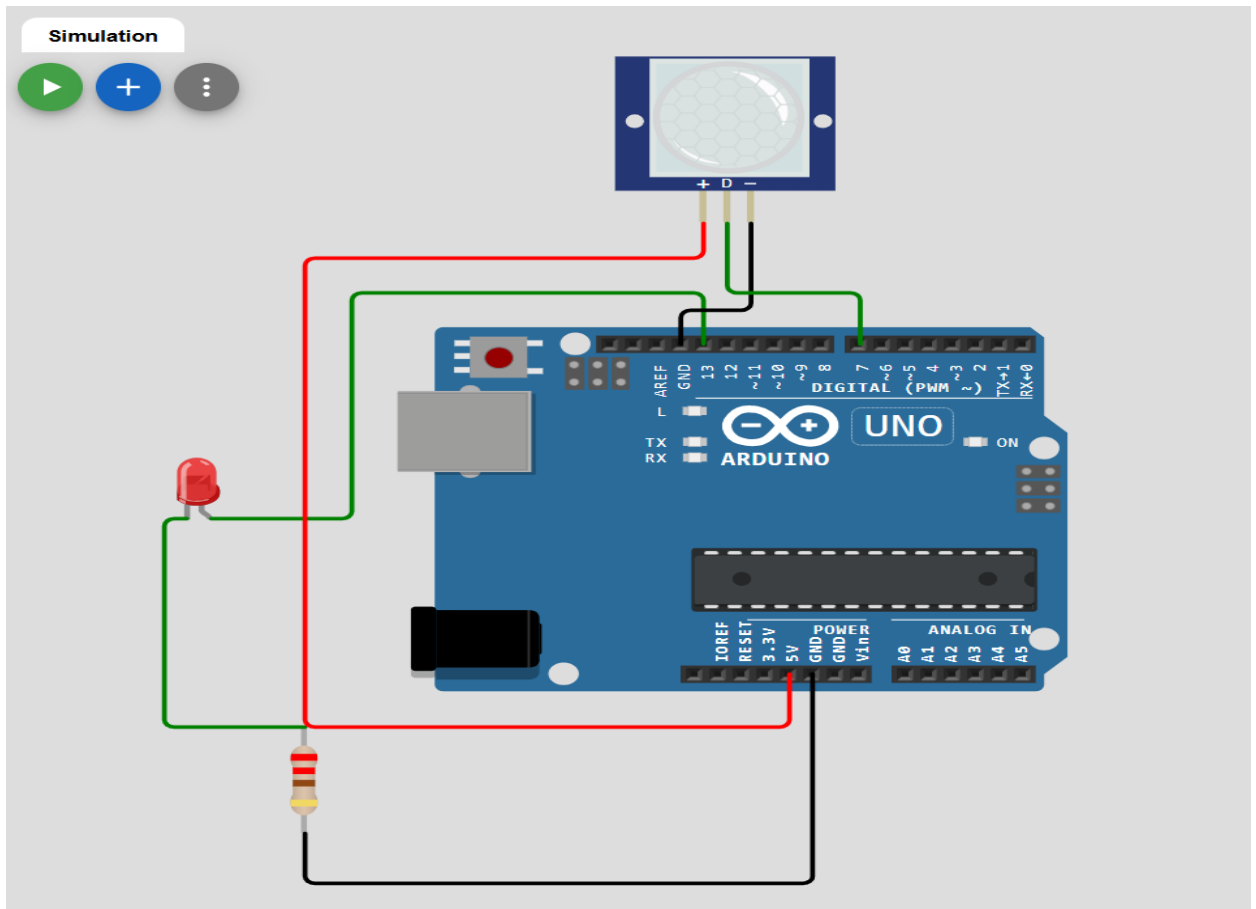
A resistor is an electronic component used to control and limit the flow of electric current in a circuit. In this project, the resistor relates to the LED to protect it from excess current that may damage the component. It helps in maintaining safe current levels and ensures proper operation of the LED. The resistor used in this project improves the reliability and safety of the circuit.



4. Working Principle

1. The PIR motion sensor continuously monitors the surrounding area for changes in infrared radiation.
2. Human movement causes changes in infrared radiation, which are detected by the PIR sensor.
3. The PIR sensor generates an output signal when motion is detected.
4. The output signal from the PIR sensor is sent to the Arduino UNO microcontroller.
5. The Arduino UNO continuously reads the input signal from the PIR sensor.
6. The microcontroller applies automation logic using IF and ELSE conditions.
7. If motion is detected, the Arduino UNO automatically turns ON the LED indicator.
8. A motion detection message is displayed in the Serial Monitor for monitoring purposes.
9. If no motion is detected, the LED remains OFF and a no-motion message is displayed.
10. The process repeats continuously to provide real-time motion detection, automation, and monitoring.

5. CIRCUIT DIAGRAM



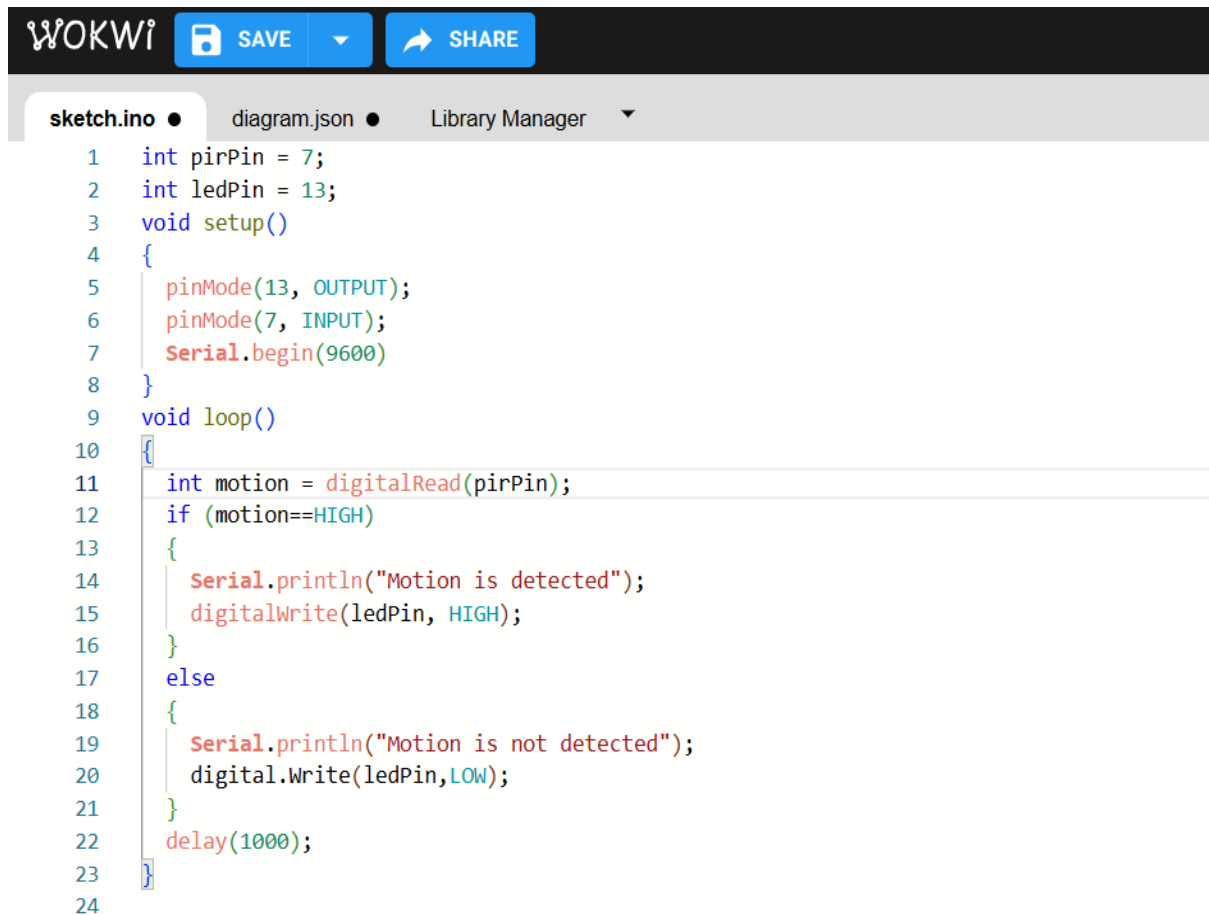
The circuit diagram of the Smart PIR Motion Sensor Security Automation System consists of an Arduino UNO, PIR motion sensor, LED, and resistor connected together to perform motion detection and automation functions. The VCC pin of the PIR sensor is connected to the 5V supply of the Arduino UNO, while the GND pin is connected to the ground terminal. The output pin of the PIR sensor is connected to a digital input pin of the Arduino UNO to send motion detection signals.

The LED is connected to a digital output pin of the Arduino UNO through a resistor, which is used to limit the current flow and protect the LED from damage. When motion is detected by the PIR sensor, the Arduino UNO processes the input signal and automatically turns ON the LED while displaying monitoring messages through the Serial Monitor. The complete circuit was designed and simulated using the Wokwi platform.

6. ALGORITHM/FLOW

1. Start the system.
2. Supply power to the Arduino UNO and initialize all components.
3. Configure the PIR sensor pin as INPUT and LED pin as OUTPUT.
4. Continuously read the signal from the PIR motion sensor.
5. Check whether motion is detected by analysing the PIR sensor output.
6. If motion is detected, send the signal to the Arduino UNO microcontroller.
7. The Arduino UNO processes the signal using IF and ELSE decision-making logic.
8. If motion is detected, turn ON the LED and display "**Motion is detected**" in the Serial Monitor.
9. If motion is not detected, turn OFF the LED and display "**Motion is not detected**" in the Serial Monitor.
10. Repeat the process continuously for real-time monitoring and automation.

7. CODE



```
WOKWI [SAVE] [SHARE]
sketch.ino • diagram.json • Library Manager
1  int pirPin = 7;
2  int ledPin = 13;
3  void setup()
4  {
5      pinMode(13, OUTPUT);
6      pinMode(7, INPUT);
7      Serial.begin(9600)
8  }
9  void loop()
10 {
11     int motion = digitalRead(pirPin);
12     if (motion==HIGH)
13     {
14         Serial.println("Motion is detected");
15         digitalWrite(ledPin, HIGH);
16     }
17     else
18     {
19         Serial.println("Motion is not detected");
20         digitalWrite(ledPin, LOW);
21     }
22     delay(1000);
23 }
24
```

CODE EXPLANATION:

1. int pirPin = 7;

This line declares an integer variable named pirPin and assigns pin number 7 to it. The PIR motion sensor output pin is connected to digital pin 7 of the Arduino UNO.

2. int ledPin = 13;

This line declares another integer variable named ledPin and assigns pin number 13 to it. The LED is connected to digital pin 13 of the Arduino UNO.

3. void setup ()

The setup () function is used to initialize the system settings. It runs only once when the Arduino is powered ON or reset.

4. pinMode(ledPin, OUTPUT);

This line sets the LED pin as an OUTPUT pin because the LED is used to display output indication.

5. pinMode(pirPin, INPUT);

This line sets the PIR sensor pin as an INPUT pin because the Arduino receives signals from the PIR sensor.

6. Serial.begin(9600);

This line starts serial communication at a baud rate of 9600 bits per second. It is used to display sensor readings in the Serial Monitor.

7. void loop ()

The loop () function contains the main program logic. It runs continuously as long as the Arduino is powered ON.

8. int motion = digitalRead(pirPin);

This line reads the signal from the PIR sensor and stores the value in the variable named motion.

9. `if (Motion==HIGH)`

This condition checks whether motion is detected. If the sensor output becomes HIGH, it indicates movement.

10. `Serial.println("Motion is detected");`

This line displays a message in the Serial Monitor indicating that motion has been detected.

11. `digitalWrite (ledPin, HIGH);`

This line turns ON the LED whenever motion is detected.

12. Else

The else block executes when no motion is detected.

13. `Serial.println("Motion is not detected");`

This line displays a message in the Serial Monitor indicating that no motion is detected.

14. `digitalWrite (ledPin, LOW);`

This line turns OFF the LED when no motion is detected.

15. `Delay (1000);`

This line creates a delay of 1000 milliseconds (1 second) before repeating the loop again.

8. APPLICATIONS

- Home security systems for detecting unauthorized movement and intrusion.
- Office security systems for monitoring restricted areas.
- Smart automation systems for automatically controlling devices based on motion detection.
- Motion-based lighting systems that turn ON lights automatically when movement is detected.
- Industrial safety systems for monitoring sensitive or restricted locations.
- Smart surveillance systems for real-time monitoring and alert generation.
- Energy-saving applications by activating devices only when human presence is detected.
- Security monitoring systems in banks, museums, shopping malls, and public places.

9. RESULT

The Smart PIR Motion Sensor Security Automation System was successfully designed and simulated using the Wokwi platform. The PIR motion sensor successfully detected human movement and sent signals to the Arduino UNO for processing. Based on the implemented automation logic, the LED automatically turned ON when motion was detected and turned OFF when no motion was present. The Serial Monitor successfully displayed real-time monitoring messages, demonstrating motion detection, automation, sensor interfacing, and embedded system concepts.

[Click here to view the simulation video](#)

10. CONCLUSION

The Smart PIR Motion Sensor Security Automation System was successfully designed and simulated using Arduino UNO and Wokwi. The system successfully detected motion, controlled the LED automatically, and displayed monitoring messages through the Serial Monitor, demonstrating automation and embedded system concepts.